Upper high school · analysis and algorithms

Bracketing a root by bisection

Worked solutions

Exercise data

For exercise 1: seek a zero of $f(x) = x^2 - 2$ on [1,2]. After two halvings, the retained interval is [1.25,1.5].

Exercise 1

Carry out a third step of the example.

The midpoint is 1.375, whose square is $1.890625 < 2$. The new interval is [1.375,1.5].

Exercise 2

Starting with width 1, how many halvings guarantee width at most $\frac{1}{1000}$?

$2^9 = 512 < 1000 \leq 1024 = 2^{10}$, so 10 halvings suffice and 9 do not.

Exercise 3

Why does the sign change of $f(x) = \frac{1}{x}$ between −1 and 1 not prove the existence of a zero?

The function is undefined at 0 and is not continuous on [−1,1]. The theorem's hypothesis fails; $\frac{1}{x}$ is never zero.

Exercise 4

A root lies in [1.4,1.5]. What maximum error is guaranteed by midpoint 1.45?

The half-width is $\frac{1.5-1.4}{2} = 0.05$.

Problem: complete reasoning

For $f(x) = x^3 - x - 1$ on [1,2], prove existence and uniqueness of a root α. Carry out three bisection steps. How many halvings guarantee that the final midpoint approximates α with error at most 0.001?

$f(1) = -1$ and $f(2) = 5$, and f is continuous. Also $f'(x) = 3x^2 - 1 \geq 2$ on [1,2], so f is strictly increasing. Midpoints 1.5, 1.25, 1.375 give [1,1.5], [1.25,1.5], [1.25,1.375] because f takes values $0.875, -0.296875$ and 0.224609375. After n halvings, midpoint error is at most $\frac{1}{2^{n+1}}$. Thus $2^{n+1} \geq 1000$, requiring $n \geq 9$.